

Lecture 12

Dirty Data

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"After analyzing all your data, I think we can safely say that none of it is useful."

Announcements

- ❖ Course project details coming soon
- ❖ Grades for labs will be up this week

Last Time

- ❖ Relational databases

- Reduces redundancy
- Enables efficiently modifying records
- Protect privacy by restricting access to some tables

Today

"Dirty data"

- ❖ Missing values
- ❖ Inconsistent formatting
- ❖ Duplicates
- ❖ Outliers

Motivation

- ❖ Thus far, we've analyzed curated datasets
- ❖ Many real-world datasets have limitations through
 - Collection procedure
 - Recording and storing the data
 - Dataset doesn't align with analysis goals
- ❖ A large part of data analysis is accounting for these sorts of limitations

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Missing Values

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 - Variables: student_id, name, age, major, gpa, grad_year
- ❖ **But** some variables are missing values
- ❖ How might this happen in practice?
 - Maybe age was never collected by the college
 - Maybe the system does not correctly calculate GPA given a pass/fail course

Representing Missing Values

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- ❖ Other ways of representing missing values:
 - String "N/A" or "Missing" for categorical variables
 - 0 or -99999 for numeric variables
- ❖ These aren't inherently bad, but it's important to document how you handled these variables
 - Part of the **data provenance**

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 - Imputation (giving them a value)
 - Typically choose the average
 - How will the value we choose affect the statistics we calculate?

Formatting

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Here are several possible answers for a student from Meadville:

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Now, imagine each group member recorded their data slightly differently!

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- ❖ How might this happen?
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 - Data may not contain unique identifiers
- ❖ Before removing duplicated records, we should ask why they're here

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- ❖ Values outside the typical range of variables in the dataset
 - Example: record with age 211
- ❖ Some outliers result from issues with the data
- ❖ Others show that there may be rare events or something interesting going on with the data
 - Example: there may just be a really tall or older student at Allegheny

Takeaways

- ❖ Data in the wild is rarely pristine
- ❖ Choices you make while handling limitations affect downstream analyses
 - Topic of this week's lab
- ❖ Important to be transparent and document your assumptions

Next Time

- ❖ Biases and Metrics
- ❖ You won't always have pristine datasets to work with!